Independent Reproduction and Convergence Analysis of the Connes–Consani–Moscovici Zeta Spectral Triple

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Abstract

We independently implement the Connes–Consani–Moscovici (CCM) operator construction [1] in Rust at arbitrary precision (MPFR/GMP) and extend the headline result from 55 to **1019.0 measured matching decimal digits** on the first Riemann zeta zero at an HP-1000 requested-precision target with a 64-bit internal guard reserve, limited by the resulting guarded working precision. We characterize the construction's convergence empirically: accuracy follows a smoothly saturating basis-size ramp that is approximately linear over the observed early range, then approaches a first-zero plateau that tracks the Weil eigenvalue depth $-\log_{10}|\varepsilon_N|$ with a slowly varying conversion gap. Along the sampled cutoff-and-basis paths, $|\varepsilon_N|$ decreases by hundreds of decimal orders over the listed prime-count doublings, although the finite data do not determine an asymptotic decay class. Across the sampled fixed-N cutoff sweep, first-zero accuracy grows monotonically with λ and closely tracks the eigenvalue depth $-\log_{10}|\varepsilon_N|$; no monotonicity claim is made outside that range. The smallest eigenvector's even symmetry, corresponding to CCM's Step 1 hypothesis, is observed at every above-floor configuration tested up to λ^2=1200, HP-2000; the unrestricted natural solve produces an essentially even eigenvector and eigenvalues agreeing with the reduced even-sector solve to hundreds of decimal digits. Remarkably, a 21×21 matrix from six primes already yields 21.585 matching digits. All results are reproducible from the accompanying source-available implementation. This work has been shared with the CCM authors and reviewed by the lead author, Professor Alain Connes.

1 Introduction

The Riemann Hypothesis (RH), asserting that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ satisfy $\Re(s) = 1/2$, remains one of the central open problems in mathematics. Among approaches toward its resolution, the Hilbert–Pólya conjecture — positing the existence of a self-adjoint operator whose spectrum encodes the imaginary parts of the nontrivial zeros — has motivated extensive work in spectral theory and noncommutative geometry.

In November 2025, Connes, Consani, and Moscovici [1] proposed the finite-dimensional operator family $D_{\log}^{(\lambda,N)}$. Its spectra exhibit striking numerical agreement with the nontrivial zeros of $\zeta(1/2+is)$ in the tested finite configurations. Their construction uses the Weil quadratic form built from the explicit formula, restricted to primes $p \leq \lambda^2$, and produces a $(2N+1) \times (2N+1)$ self-adjoint matrix whose smallest eigenvalue ε_N is observed to decrease over the tested families as λ and N increase. They reported accuracy of 2.5×10^{-55} on the first zero at $(\lambda^2{=}13, N{=}120)$ with 200-digit working precision. Subsequent independent work by Groskin reproduced the high-precision zero approximation from the truncated Weil form across multiple cutoffs [2].

In this paper we:

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1. Independently reproduce the CCM headline result on a Rust + MPFR/GMP software stack;

2. Extend the construction to (λ^2=1000, N=800) at an HP-1000 requested-precision target with 64 guard bits, achieving **1019.0 measured matching digits** on the first Riemann zero, limited by guarded working precision rather than by the construction's intrinsic accuracy;

3. Systematically characterize the convergence behavior as a function of both N (basis size) and λ (prime cutoff);

4. Identify structural properties of the construction not discussed in [1], including empirical confirmation that the smallest eigenvector's even symmetry (Step 1 hypothesis) holds at every configuration tested above the precision floor; we explain the mechanism by which insufficient working precision produces erroneous mixed-symmetry artifacts when ε_N falls below the precision floor.

All computations are reproducible from the source-available implementation at `https://github.com/TeamXcelerator/ccm-reproduction-and-convergence`.

2 Mathematical Setup

We briefly recall the CCM construction, following the notation of [1].

2.1 The Weil Quadratic Form

Let $\lambda > 1$ and $L = 2\ln\lambda$. The Hilbert space is $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], du/u)$ with orthonormal basis

$$V_n(u) = \frac{1}{\sqrt{L}} e^{2\pi i n \ln(\lambda u)/L}, \quad n \in \{-N, \dots, N\}.$$

The Weil quadratic form QW_λ is a real symmetric $(2N{+}1) \times (2N{+}1)$ matrix with entries

$$\tau_{n,m} = W_{0,2}(V_n, V_m) - W_R(V_n, V_m) - \sum_{\substack{k=p^j \\ k \le \lambda^2}} \frac{\log p}{\sqrt{k}} q(V_n, V_m)(\log k),$$

where $W_{0,2}$ is a closed-form rank-two pole correction, W_R is the archimedean (gamma-factor) contribution computed via Gauss–Legendre quadrature, and the prime-power sum encodes the von Mangoldt function $\Lambda(k) = \log p$ for $k = p^j$.

2.2 From Eigenvector to Spectral Ordinates

Let ε_N denote the smallest eigenvalue of τ and $\xi = (\xi_{-N}, \dots, \xi_N)$ the corresponding eigenvector. In the reproduction pipeline we analyze the even representative ($\xi_{-j} = \xi_j$) and normalize it so that $\sum_j \xi_j = \sqrt{L}$. Define the secular rational function

$$R(s) = \xi_0 + 2s \sum_{j=1}^{N} \frac{\xi_j}{s - j^2}.$$

Let $s_1 < \cdots < s_M$ denote its positive real roots, indexed in increasing order. The corresponding positive spectral ordinates are

$$\nu_r = \frac{2\pi}{L}\sqrt{s_r}.$$

The root ordinal r is not the pole-interval index j: many intervals $(j^2, (j+1)^2)$ can be empty, and at small N some roots matching low zeta ordinates can lie above the largest pole N^2. Under the CCM construction, the ν_r approximate the positive ordinates γ_r defined by $\zeta(1/2+i\gamma_r) = 0$.

All tables reporting "matching digits" use the relative metric

$$D_r(\lambda^2, N, P) := -\log_{10}\left(\frac{|\nu_r - \gamma_r|}{|\gamma_r|}\right). \tag{1}$$

Section 4.1 also reports absolute error to compare directly with the original CCM headline result.

3 Implementation

Our implementation uses the Xcelerator Toolkit,[1] a publicly available library of high-precision spectral methods for analytic number theory, pinned to a tagged release for reproducibility. The toolkit provides the core numerical primitives (Gauss–Legendre quadrature with disk-cached HP nodes, LU-based inverse iteration, Newton refinement) and the CCM-specific construction (Weil-form matrix assembly, eigenvector extraction, rational-function root-finding). The paper-specific repository contains only the CLI harness and reproduction scripts; all mathematical code is shared via the toolkit.

The toolkit is built on:

- **Rust** (edition 2021) for the main codebase;
- **rug** (v1.30, wrapping MPFR/GMP) for arbitrary-precision floating-point arithmetic;
- **nalgebra** (v0.33.3) for the f64 fast-path eigendecomposition (used only by the optional `--f64-only` mode for smoke tests; the HP publication path uses the toolkit's own HP symmetric eigendecomposition in `xc-numerics::eigen`, verified at HP-1000 against a PARI/GP 2000-digit reference [3]);
- **rayon** (v1.12.0 in the locked dependency graph) for parallel matrix assembly, Householder reduction, LU factorization, and eigenvalue-per-seed refinement.

Key algorithmic choices:

1. **Parity-aware eigenstate solve:** Paper claims use the direct reduced even-sector matrix. The default automatic solver first reuses an exact compatible eigenstate when available; on a cache miss it applies a retained-factor shift-invert Krylov/Rayleigh–Ritz method, optionally initialized by a compatible lower-N cached state. A residual-verified LU inverse-iteration route remains the conservative fallback when the targeted Ritz value is not isolated unambiguously. The unrestricted natural and conditional-projection routes use the full-space inverse-iteration implementation. Parity policy and solver route are part of artifact identity, so numerically equivalent states from different operation sequences cannot be confused in cache reuse.
2. **Halley's method for $R(s)$ zeros:** The rational function $R(s) = \xi_0 + 2s\sum_{j=1}^{N} \xi_j/(s-j^2)$ has analytically available first and second derivatives, enabling Halley's method (cubic convergence) as the default root-finder. The paper-claim scripts seed refinement from a toolkit-owned reference dataset whose zeros were independently computed to 2,500 decimal digits using rigorous Arb interval arithmetic [4]; their leading 1,000 digits were cross-validated against Odlyzko's tabulation [5]. The reference ordinate selects which finite secular root is refined; it is not substituted into the result. An accepted iterate must

[1] `https://github.com/TeamXcelerator/xcelerator-toolkit`

satisfy the finite secular equation and the solver's convergence tests. Independent finite-source discovery is also available for unseeded research runs. The point solver distinguishes four outcomes: `Converged` (the requested correction target was met), `Stagnated` (a finite approximation stopped improving), `Approximate` (the iteration limit was reached with a finite value), and `Failed` (no usable finite value). Computed workflows retain finite nonconverged values with diagnostics; they are not mislabeled as converged or certified.

3. **Gauss–Legendre quadrature with disk cache:** GL nodes and weights at a given quadrature order and bit precision are computed once and cached for reuse across runs.

3.1 Reproducibility Infrastructure

The toolkit manages typed artifacts for quadrature rules, the archimedean and prime components, τ matrices, parity reductions, factorizations, Weil states, secular sources, root refinements, and validation evidence. Each artifact has a content-bound manifest whose semantic identity includes the numerical parameters, parity policy, algorithm route, precision contract, and dependency identities that can affect its value. Loads replay family-specific structural checks; incompatible or obsolete artifacts are rejected rather than silently reused.

Artifacts are stored first in a per-user local cache. Depending on the selected cache policy, compatible objects may also be resolved from managed public or authorized private shards; an ordinary user needs no credentials and computes locally on a miss. Publication is a separate author-only operation. Consequently, reproducibility does not depend on a particular remote fixture being present: a cache hit accelerates the run, while a miss executes the same content-identified numerical route.

4 Results

4.1 Reproduction of the CCM Headline

At ($\lambda^2{=}13, N{=}120,$ 200-digit precision), our first computed spectral ordinate is

$$\nu_1 = 14.134725141734693790457251983562470270784257\ldots$$

(displayed truncated to 42 digits; the full HP-1000 result has $D_1 = 55.764$ under (1), see Section 4.3) with absolute error 2.4363×10^{-55} against the independently computed 2,500-digit reference. This matches the CCM paper's reported 2.5×10^{-55} to within rounding. The result is unchanged at an HP-1000 requested-precision target, confirming that the observed accuracy tracks $\varepsilon_N \approx 3.48399 \times 10^{-59}$ rather than being limited by arithmetic precision.

4.2 Extension to 1019 Measured Digits

Table 1: Measured matching digits across configurations, all at an HP-1000 requested-precision target with a 64-bit internal guard reserve. At λ^2=1000 the reported agreement saturates guarded working precision; the construction's intrinsic ceiling is not resolved by this run.

Configuration	First-zero digits	20th-zero digits	Requested precision
$\lambda^2{=}13$, $N{=}120$	55.764	25.339	HP-1000
$\lambda^2{=}100$, $N{=}500$	460.09	422.44	HP-1000
$\lambda^2{=}1000$, $N{=}800$	**1019.0**	**1020.3**	HP-1000

Here HP-1000 denotes the requested accuracy contract, not the raw MPFR allocation. The implementation converts 1000 decimal digits to 3322 requested bits and appends 64 guard bits, giving 3386 working bits, or approximately 1019.3 decimal digits. Root acceptance is tested against the 3322-bit requested target; the guard reserve protects that target from cancellation and accumulated rounding error. The matching-digits diagnostic, however, compares the final values at the full guarded precision, so a converged Halley iteration can expose agreement within the reserve: 1019.0 digits for the first zero in the latest run. The 20th-zero relative metric reaches 1020.3 because it is normalized by $|\gamma_{20}| > 1$, which can place it slightly above the raw decimal-digit equivalent of the bit allocation. These guard-space measurements do not redefine HP-1000 as a 1019-digit requested calculation; they report the additional agreement preserved by the internal working precision.

At configurations where ε_N is smaller than the working-precision floor (i.e., the construction's intrinsic accuracy exceeds what the arithmetic can resolve), raising the working precision reveals additional matching digits approximately one-for-one. In general, accuracy is controlled jointly by λ (via ε_N), N (basis completeness), and working precision; all three must be scaled together to sharpen beyond a given level.

4.3 Convergence in N (Basis Size)

Table 2: Matching digits on the first five zeros vs. N at λ^2=13. Values are identical at HP-200 and HP-1000 across the entire sweep (see commentary below); the column shown is from the HP-1000 run.

N	Zero 1	Zero 2	Zero 3	Zero 4	Zero 5
10	21.585	16.335	11.856	5.1626	3.4915
20	35.795	32.261	30.011	26.504	24.870
30	44.506	41.267	39.262	36.208	34.844
40	50.294	47.152	45.225	42.304	41.015
50	53.827	50.721	48.822	45.948	44.685
60	55.299	52.203	50.312	47.451	46.195
80	55.654	52.560	50.670	47.811	46.556
100	55.735	52.641	50.751	47.892	46.637
120	55.764	52.670	50.779	47.921	46.666

The sweep shows a smoothly saturating ramp with two useful empirical regimes:

1. **Rapid pre-saturation growth** ($N = 10$ to $N \approx 60$): matching digits are approximately linear over the early part of the sweep, while the local rate $\Delta(\text{digits})/\Delta N$ decreases smoothly from ∼1.4 at N=10→20 to ∼0.15 at N=50→60. The average ∼0.96 digits per unit N over $N \in [10, 40]$ is a local summary, not a global rate law.

2. **Near-plateau behavior** ($N > 60$): the first-zero accuracy approaches 55.764 digits by $N = 120$. At this configuration,

$$D_\varepsilon := -\log_{10} |\varepsilon_N| = 58.458,$$

so the first-zero plateau lies 2.694 digits below the eigenvalue depth. We therefore call it the root-accuracy plateau associated with ε_N, not the eigenvalue depth itself.

The precise local slope and plateau location are properties of the sampled (λ, N) regime. Here $\lambda^2 = 13$ and $N \in \{10, 20, \ldots, 120\}$; the paper does not assert the same numerical break point at other cutoffs.

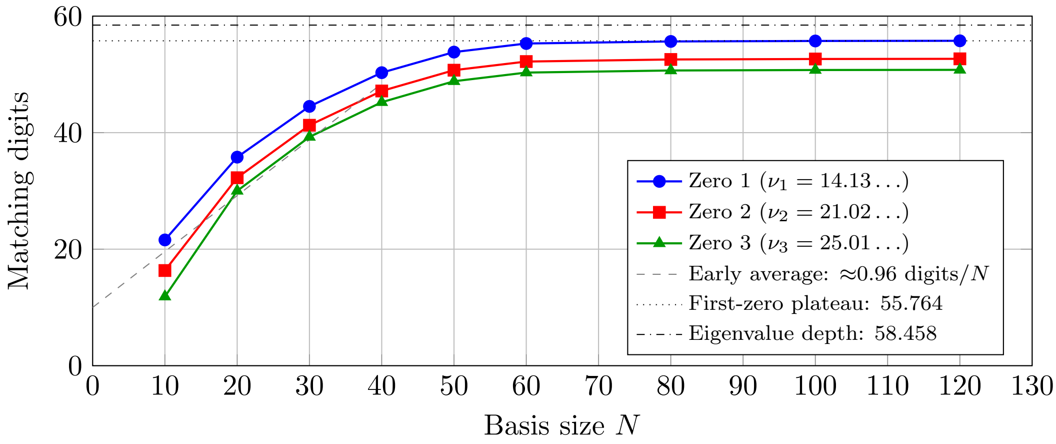

Figure 1: Convergence of matching digits vs. basis size N at $\lambda^2 = 13$, HP-1000. The two horizontal lines distinguish the first-zero plateau from the deeper Weil-eigenvalue scale.

Precision comparison. At every N in this sweep, HP-200 and HP-1000 produce identical matching digits. The observed plateau is therefore intrinsic to the finite configuration rather than imposed by the arithmetic precision; Sections 4.6 and 4.7 describe the corresponding floor mechanism in other regimes.

4.4 The 21-Digit Result from a 21×21 Matrix

At $N = 10$ (matrix size 21×21), with only six primes ($\{2, 3, 5, 7, 11, 13\}$, corresponding to nine prime powers entering the Weil form), the first Riemann zero is matched to **21.585 decimal digits**. This demonstrates notable numerical compression: more than 21 matching digits from a 21×21 matrix.

This small configuration also illustrates that the lattice edge is not a hard root cutoff. Here

$$T_{\max} = \frac{2\pi N}{L} \approx 24.50,$$

yet the third through fifth zeta ordinates $\gamma_3, \gamma_4, \gamma_5 \approx 25.01, 30.42, 32.94$ still have accurate finite matches, and the corresponding secular roots lie above the largest pole $N^2 = 100$. These beyond-band matches are empirical features of the finite secular equation; no general localization claim is inferred from this example.

4.5 Rapid Empirical Change of ε_N

Table 3: Selected finite-configuration values of the smallest Weil eigenvalue. The explicit N column is essential because ε_N depends on both cutoff and basis size.

λ^2	N	Primes $\leq \lambda^2$	ε_N	$-\log_{10} \lvert\varepsilon_N\rvert$
13	120	6	3.48399×10^{-59}	58.458
100	500	25	9.56481×10^{-464}	463.02
1000	800	168	$3.92215 \times 10^{-1264}$	1263.41

These values show a rapid increase in eigenvalue depth along the listed finite-configuration path. They should not be read as a prime-count-only law: the three rows use different basis sizes and sit at different positions on the finite-N ramp. The next table gives a more systematic diagonal path with $N/\sqrt{\lambda^2} \approx 28$.

Table 4: Smallest Weil eigenvalue ε_N above the working-precision floor, with N scaled at the fixed ratio $N/\sqrt{\lambda^2} \approx 28$. HP-1500 for $\lambda^2 = 500$–800 and HP-2000 for $\lambda^2 = 1000$–1200. The final column gives the change in eigenvalue-depth digits, normalized per doubling of prime count and computed between consecutive rows; it is illustrative rather than a universal rate.

λ^2	N	Primes $\leq \lambda^2$	ε_N	$-\log_{10} \lvert\varepsilon_N\rvert$	Depth/doubling
500	630	95	1.09117×10^{-929}	928.96	—
600	690	109	$2.22999 \times 10^{-1029}$	1028.65	502.7
700	740	125	$9.45898 \times 10^{-1116}$	1115.02	437.1
800	790	139	$1.35946 \times 10^{-1199}$	1198.87	547.5
1000	890	168	$1.52095 \times 10^{-1362}$	1361.82	596.1
1200	970	196	$6.79867 \times 10^{-1499}$	1498.17	613.1

The depth grows rapidly throughout this sampled family. The per-doubling numbers depend on the parameterization and, critically, on the chosen basis path $N(\lambda)$. A different scaling of N produces different values. Six finite samples cannot establish whether the asymptotic decay is polynomial, exponential, or super-exponential. A rigorous two-parameter law for $\varepsilon_N(\lambda, N)$ remains open.

4.6 Observed Monotone Convergence in λ at Fixed N

At fixed $N = 120$, first-zero accuracy grows monotonically across the sampled cutoff range and closely tracks the eigenvalue depth

$$D_\varepsilon(\lambda) := -\log_{10} |\varepsilon_N(\lambda)|.$$

The two quantities are not identical: their difference is the finite eigenvalue-to-root conversion gap.

Table 5: First-zero accuracy vs. λ^2 at $N = 120$, comparing HP-200 and HP-1000 requested precision. The final column shows the slowly varying conversion gap $C_{\text{conv}} = D_\varepsilon - D_1$.

λ^2	Primes	ε_N at HP-1000	HP-200	HP-1000	C_{conv}
13	6	3.48399×10^{-59}	55.764	55.764	2.694
20	8	2.50471×10^{-96}	92.817	92.817	2.784
30	10	2.19404×10^{-135}	131.80	131.80	2.859
50	15	6.89051×10^{-174}	170.19	170.19	2.972
100	25	3.48656×10^{-215}	211.29	211.29	3.168

For HP-200, the implementation converts the 200-decimal-digit request to 665 bits and adds 64 guard bits, yielding 729 working bits, or approximately 219.4 decimal digits. Root acceptance is still measured against the 665-bit requested target. The 211.29-digit result at $\lambda^2 = 100$ therefore uses part of the guard reserve without changing the nominal HP-200 accuracy contract. Diagnostics closer to the floor remain precision-sensitive: HP-200 gives $\varepsilon_N = 3.48676 \times 10^{-215}$, compared with 3.48656×10^{-215} at HP-1000. The corresponding complete-QR GapLog values are 4.246343 and 4.246284.

Two empirical regularities emerge:

1. **Observed monotone growth over the sampled range.** First-zero accuracy increases at every listed cutoff. No fixed-N monotonicity claim is made outside this finite sweep.

2. **ε_N-tracked root accuracy.** The first-zero digits follow D_ε with a conversion gap that grows slowly from about 2.7 to 3.2 digits across the table.

The cutoff dependence is not featureless. Each prime-power threshold changes the finite matrix path and produces a localized slope kink, precisely the event structure isolated by Groskin [6]. In sampled accuracy and eigenvalue curves these events appear as fine-scale perturbations on an otherwise smooth envelope, rather than as a separate staircase law.

For practitioners, increasing the cutoff is useful only when basis size and working precision are scaled with the desired target. The present sweep establishes monotone growth only for $13 \le \lambda^2 \le 100$ at $N = 120$; it does not assert that increasing λ alone is sufficient outside that range.

4.7 Critical N at Fixed Working Precision

At fixed working precision, the construction reaches an apparent ceiling beyond which the computed eigenstate or root refinement can become floor-limited or stagnate. This ceiling is precision-dependent, not a property of the construction itself.

Table 6: Convergence at large N at HP-1000. Configurations that plateau at HP-200 continue to converge cleanly at HP-1000.

Configuration	ε_N at HP-1000	First-zero digits	5th-zero digits
λ^2=50, N=200	5.69795×10^{-221}	217.34	207.24
λ^2=50, N=250	4.51247×10^{-239}	235.45	225.42
λ^2=100, N=300	4.68074×10^{-368}	364.37	354.03
λ^2=100, N=400	6.12702×10^{-423}	419.28	409.04

In a locally approximately linear part of the mode ramp, let r denote the observed gain in matching digits per added basis mode. The quantity

$$N_{\text{crit}} \approx \frac{P_{\text{work}}}{r}$$

is then an estimate of the basis size at which working precision becomes binding; it is not itself an accuracy ceiling. At $\lambda^2 = 13$ the early-range average is $r \approx 0.96$ digits per mode, so 200 working digits suggest $N_{\text{crit}} \approx 210$. The estimate is local and configuration-dependent because the ramp slope decreases with N.

The actual measurement ceiling in digits is set by the available working precision, subject to conditioning and guard-space behavior. Configurations that appear to plateau at HP-200 continue to improve at HP-1000, as Table 6 shows. A practical calculation should therefore scale N and precision together rather than infer an intrinsic plateau from a single-precision run.

4.8 Even-Symmetry of the Smallest Eigenvector

The CCM proof's Step 1 requires the smallest eigenvector ξ to be even under the involution induced by $u \mapsto 1/u$. Let J denote the corresponding coefficient reversal, $(J\xi)_j = \xi_{-j}$. We measure the natural evenness $\|\xi - J\xi\|/\|\xi\|$ of the unforced smallest eigenvector at working precisions sufficient to resolve ε_N above the floor:

The λ^2=100, N=500 row was remeasured with the current 64-bit guard reserve; the earlier 16-guard-bit calculation gave 5.102×10^{-548}. The additional 48 working bits supply approximately $48 \log_{10} 2 = 14.45$ decimal digits of arithmetic headroom, consistent with the roughly 15-order reduction to 6.665×10^{-563}. This change lowers the residual symmetry noise without changing the smallest eigenvalue, its sign, or the "essentially even" conclusion.

The λ^2=1000 rows show the same guard-reserve effect. At N=800 the deviation falls from the earlier 2.941×10^{-748} to 7.634×10^{-763}, and at N=890 from 6.279×10^{-650} to 1.563×10^{-664}. Both improvements are approximately 14.6 decimal orders, consistent with the added 48 guard

Table 7: Symmetry properties of the natural (unforced) smallest eigenvector, each measured at working precision sufficient to resolve ε_N above the floor. At every configuration tested, the natural eigenvector is essentially even and the natural and even-sector eigenvalues agree to hundreds of decimal digits.

λ^2	N	Precision	Evenness deviation	Eigenvalue rel. diff.	Verdict
13	120	HP-1000	2.126×10^{-966}	1.107×10^{-962}	essentially even
100	500	HP-1000	6.665×10^{-563}	0 (at working precision)	essentially even
1000	800	HP-2000	7.634×10^{-763}	0 (at working precision)	essentially even
1000	890	HP-2000	1.563×10^{-664}	0 (at working precision)	essentially even
1200	970	HP-2000	2.948×10^{-528}	7.927×10^{-526}	essentially even

bits; the natural and even-sector eigenvalues remain numerically equivalent at the reported precision.

The λ^2=1200, N=970 row was likewise remeasured with the 64-bit guard reserve. Its evenness deviation falls from 2.927×10^{-514} to 2.948×10^{-528}, an improvement of approximately 14.0 decimal orders. The natural and even-sector eigenvalues are numerically equivalent at the attainable relative precision, agreeing to approximately 525 decimal digits with relative difference 7.927×10^{-526}. Their full 6708-bit representations need not be bit-identical: the natural full-space solve and the reduced even-sector solve follow different high-precision operation sequences, so roundoff at the working-precision floor need not produce identical terminal bit patterns. Moreover, 6708 bits provide approximately 2019 decimal digits, while $|\varepsilon_N| \approx 6.799 \times 10^{-1499}$ consumes roughly 1499 orders of magnitude; the resulting $\sim$520 relative-digit headroom is consistent with the observed 525-digit agreement. Thus the nonzero difference reflects the arithmetic floor rather than a mathematical discrepancy between the two eigenvalues.

At every configuration tested above the precision floor — including $\lambda^2 = 1200$, far beyond the range previously reported — the natural smallest eigenvector is even to hundreds of digits of precision, and the natural and even-sector solves converge to numerically indistinguishable eigenvalues at the reported accuracy. In the current λ^2=13, N=120, HP-1000 reproduction, performed at 3386-bit working precision, their relative difference is 1.107×10^{-962}. CCM's Step 1 hypothesis holds empirically at every tested configuration. We conjecture it holds universally: the smallest eigenspace of QW_λ is contained in the even sector for all (λ, N).

The precision-floor artifact. When $|\log_{10} \varepsilon_N|$ approaches or exceeds the working precision in digits, the eigenvalue sinks below the arithmetic's resolving power. In this regime the computed eigenvector is under-resolved and becomes sensitive to rounding within numerically indistinguishable directions. The result can *appear* non-even (deviation $\sim O(1)$), while the eigenvalues can *appear* to split between the natural and even-sector paths. Under the earlier guard policy, the HP-1000 $\lambda^2 = 1000$ configuration ($|\log_{10} \varepsilon_N| \approx 1264$, well past the available arithmetic floor) exhibits this artifact: deviation 1.87, eigenvalue split 24%. Increasing the requested precision to HP-2000 under the same policy at the *identical* (λ^2=1000, N=800) configuration collapses the apparent breakdown entirely: deviation 2.94×10^{-748}, with eigenvalues numerically equivalent at the reported precision. Thus the only variable in that controlled comparison is requested precision. The latest 64-guard-bit HP-2000 run uses 6708 working bits (approximately 2019.3 decimal digits, a floor ratio of approximately 1.60) and reduces the residual symmetry noise further to 7.634×10^{-763}, again with numerically equivalent natural and even-sector eigenvalues at the reported precision.

The even-sector restriction is empirically unnecessary. To further test whether the eigenvector is naturally even, we ran the full CCM pipeline (eigenstate solve $\to$ eigenvector

$\to R(s)$ zero-finding $\to$ Riemann-zero matching digits) with the unrestricted natural full-space policy at configurations spanning $\lambda^2 = 13$–1000, $N = 10$–800, and HP-200 through HP-2000. At every tested above-floor point, the natural path produces eigenvalues and Riemann-zero approximations numerically indistinguishable from the reduced even-sector path at the reported accuracy. The direct restriction is therefore not needed to obtain the reported values at these points, although it remains the faster paper-reproduction default.

5 Discussion

5.1 Where the Construction's Power Lies

The CCM construction's measured accuracy is controlled jointly by three caller-selected parameters:

- λ sets the arithmetic cutoff and changes the intrinsic finite-configuration depth;
- N controls progress along the finite-basis ramp toward that depth;
- working precision determines how much of the finite result can be resolved without floor artifacts.

The sampled path $N/\sqrt{\lambda^2} \approx 25$–30 was sufficient to reach the available HP-1000 and HP-2000 precision targets in the large-cutoff runs. It is not a universal rule for reaching the intrinsic finite-cutoff plateau: Tables 4 and 6 still show substantial gains from additional modes beyond that ratio. A companion convergence-law study [7] examines the broader coordinate $N/(\lambda^2 \ln \lambda^2)$; the present reproduction paper does not assume that model. In practice, λ, N, and precision should be chosen together for the desired accuracy target.

5.2 The Prime-Weight Profile

Each prime power $k = p^j$ contributes weight $\Lambda(k)/\sqrt{k} = \log(p)/\sqrt{k}$ to the Weil form. For primes ($j = 1$), this is $\log(p)/\sqrt{p}$, whose derivative vanishes at $p = e^2 \approx 7.3891$ with maximum value $2/e \approx 0.73576$.

Table 8: Individual prime contributions $\log(p)/\sqrt{p}$ for the primes captured by $\lambda^2 = 13$. The maximum value $2/e$ is attained at the (non-prime) point $p = e^2 \approx 7.3891$; the integer prime 7 achieves 99.96% of this maximum.

p	$\log(p)/\sqrt{p}$	% of $2/e$
2	0.49013	66.62%
3	0.63428	86.21%
5	0.71976	97.83%
7	**0.73548**	**99.96%**
11	0.72299	98.26%
13	0.71139	96.69%
max $= 2/e$ at $p = e^2$		100%

The following observation describes a property of the resulting finite prime set; it does not imply that the original cutoff was selected for this reason.

The CCM paper's choice of $\lambda^2 = 13$ has the useful consequence that it captures the peak prime $p = 7$ to within 0.04% of the maximum value and includes both adjacent primes ($p = 5$ at 97.8% of the maximum and $p = 11$ at 98.3%) that lie within $\sim 2\%$ of the peak. The choice covers the entire near-peak cluster of primes.

Past the peak, $\log(p)/\sqrt{p}$ decays as a power law with a logarithmic factor (0.46052 at $p = 100$, 0.21844 at $p = 1000$, 0.092103 at $p = 10000$). Thus later primes still have visible individual weights at the finite cutoffs studied here. Their cumulative effect is consistent with the measured reduction in $|\varepsilon_N|$, but the scalar weight profile alone does not derive the eigenvalue-decay law; matrix interactions and the simultaneous scaling of N remain essential.

5.3 Implications for the Proof Strategy

Our findings have implications for the CCM proof strategy:

1. The even symmetry of the smallest eigenvector (Table 7) is observed at every configuration tested above the precision floor, up to $\lambda^2 = 1200$ at HP-2000. We conjecture that the smallest eigenspace is always contained in the even sector. The direct even-sector restriction is empirically unnecessary at the tested above-floor points.

2. The basis sweep in Table 2 shows diminishing returns only after the finite-cutoff root plateau is approached. Before that plateau, Tables 4 and 6 show that substantial gains from increasing N remain possible.

3. The rapid measured change of ε_N along the sampled (λ, N) paths supplies empirical convergence evidence, but both a rigorous two-parameter law and the asymptotic decay class remain open.

6 Conclusion

We have independently reproduced and significantly extended the CCM zeta spectral triple construction, achieving 1019.0 measured matching digits on the first Riemann zero at an HP-1000 requested target with a 64-bit guard reserve, limited by guarded working precision. The key findings are:

- A smoothly saturating basis-size ramp that is approximately linear over the observed early range.

- Rapid empirical growth of the Weil-eigenvalue depth along the tested cutoff-and-basis paths, without a claim that prime count alone determines the rate.

- Observed monotone growth of first-zero accuracy across the sampled fixed-N cutoff sweep, with a slowly varying conversion gap between root accuracy and $-\log_{10}|\varepsilon_N|$.

- A precision-dependent measurement ceiling at finite N; the critical basis size at which precision becomes binding is distinct from the digit ceiling itself.

- Even symmetry of the smallest eigenvector at every tested above-floor configuration through $\lambda^2 = 1200$, HP-2000; apparent symmetry breakdown in floor-limited data is a precision artifact.

- Notable low-dimensional compression: a 21×21 matrix from six primes already produces 21.585 matching digits.

Together these results provide a comprehensive empirical foundation for future theoretical work on the construction.

The observed compression suggests algebraic structure in the Weil quadratic form that is not yet theoretically understood. Explaining why this mechanism produces Riemann zeros with such precision remains an open problem whose resolution could advance the spectral approach to the Riemann Hypothesis.

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Statement on the Use of AI Tools

In accordance with current publication practices concerning authors' use of generative AI tools, the author discloses that Anthropic's Claude and OpenAI's Codex served as aids in preparing the exposition and the Rust implementation code, to specifications set by the author. AI assistance was used to improve prose formatting and readability, gather research material, propose candidate arguments, and validate calculations; all such output was reviewed by the author. The author originated, directed, and conducted every computational experiment presented here. The mathematical development was led by the author with assistance from AI. The author bears full responsibility for the contents of this manuscript.

Data and Code Availability

The complete paper-specific source code and reproduction scripts are available at:
`https://github.com/TeamXcelerator/ccm-reproduction-and-convergence`
The shared mathematical library and content-bound reference-zero dataset are supplied by the Xcelerator Toolkit at:
`https://github.com/TeamXcelerator/xcelerator-toolkit`

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